

Last login: Tue Jun 16 14:12:44 on ttys000
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch
psql (18.4)
Type "help" for help.

```
tpch=# SET ROLE alice;
SET
tpch=> select
  o_orderpriority,
  count(*) as order_count
from
  orders
where
  o_orderdate >= date '1994-01-01'
  and o_orderdate < date '1994-01-01' + interval '3' month
  and exists (
    select
      *
    from
      lineitem
    where
      l_orderkey = o_orderkey
      and l_commitdate < l_receiptdate
  )
group by
  o_orderpriority
order by
  o_orderpriority;
WARNING:  Log 1: Planner Input:
DETAIL:  {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTTRY
:alias <>
:eref
  {ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 16561
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals (
  {SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
  {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTTRY
:alias
  {ALIAS
:aliasname c
:colnames <>
}
:eref
  {ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")
}
:rtekind 0
:relid 16546
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
  {RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointree
  {FROMEXPR
:fromList (
  {RANGETBLREF
:rtable 1
}
)
}
:quals
  {BOOLEXP
:boolop and
:args (
  {OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
:varno 1
:varattno 1
}
```

```

        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location -1
    }
    {VAR
        :varno 1
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 1
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 2
        :location -1
    }
}
:location -1
}
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
    {VAR
        :varno 1
        :varattno 6
        :vartype 701
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 6
        :location -1
    }
    {FUNCEXPR
        :funcid 316
        :funcresulttype 701
        :funcset false
        :funcvariadic false
        :funcformat 2
        :funccollid 0
        :inputcollid 0
        :args (
            {CONST
                :consttype 23
                :consttypmod -1
                :constcollid 0
                :constlen 4
                :constbyval true
                :constisnull false
                :location -1
                :constvalue 4 [ -128 19 0 0 0 0 0 0 ]
            }
        )
        :location -1
    }
}
:location -1
}
}
:location -1
}
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
        :expr
        {CONST
            :consttype 23
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull false
            :location -1
            :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
        }
        :resno 1
        :resname ?column?
        :resortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
    }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
{RANGETBLENTY
:alias <>
:eref
    {ALIAS
        :aliasname *GROUP*
        :colnames ("o_orderpriority")
    }
}
:rtekind 9
:groupexprs (
    {VAR
        :varno 1
        :varattno 6
        :vartype 25
        :vartypmod -1
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 6
        :location 11
    }
)

```

```

    )
    :lateral false
    :inFromCl false
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
  :relid 16561
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 0 12 13)
  :insertedCols (b)
  :updatedCols (b)
  }
)
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
  :rtindex 1
  }
)
}
:quals
{BOOLEXP
:boolop and
:args (
  {OPEXP
  :opno 1098
  :opfuncid 1090
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 1
    :varattno 5
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 5
    :location 82
    }
    {CONST
    :consttype 1082
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 102
    :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
    }
  )
  :location 94
  }
  {OPEXP
  :opno 2345
  :opfuncid 2338
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 1
    :varattno 5
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 5
    :location 123
    }
    {OPEXP
    :opno 1076
    :opfuncid 2071
    :opresulttype 1114
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {CONST
      :consttype 1082
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location 142
      :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
      }
      {CONST
      :consttype 1106
      :consttypmod 196607
      :constcollid 0
      :constlen 16
      :constbyval false
      :constisnull false
      :location 166
      :constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
      }
    )
  )
  :location 155
  }
)
:location 135
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
  :alias <>
  :eref
  {ALIAS
  :aliasname lineitem
  :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu

```

```

        mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
        "l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate"
        "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
        t")
    }
    :rtekind 0
    :relid 16566
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
23)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 282
}
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 295
}
)
:location 293
}
{OPEXP
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 322
}
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 337
}
)
:location 335
}
)
:location 318
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 220
}
}
:resno 1
:resname l_orderkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1

```

```

:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 220
}
:resno 2
:resname 1_partkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 220
}
}
:resno 3
:resname 1_supplekey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 220
}
}
:resno 4
:resname 1_linenummer
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 220
}
}
:resno 5
:resname 1_quantity
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 220
}
}
:resno 6
:resname 1_extendedprice
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 220
}
}
:resno 7
:resname 1_discount
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 220
}
}
:resno 8
:resname 1_tax
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 8

```

```

:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 220
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location 220
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 220
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 220
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 220
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location 220
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location 220
}

```

```

    }
    :resno 15
    :resname l_shipmode
    :ressortgroupref 0
    :resorigtbl 16566
    :resorigcol 15
    :resjunk false
  }
  {TARGETENTRY
  :expr
  {VAR
  :varno 1
  :varattno 16
  :vartype 25
  :vartypmod -1
  :varcollid 100
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 1
  :varattnosyn 16
  :location 220
  }
  :resno 16
  :resname l_comment
  :ressortgroupref 0
  :resorigtbl 16566
  :resorigcol 16
  :resjunk false
  }
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 184
}
:location 119
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList {
  {TARGETENTRY
  :expr
  {VAR
  :varno 2
  :varattno 1
  :vartype 25
  :vartypmod -1
  :varcollid 100
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 2
  :varattnosyn 1
  :location -1
  }
  :resno 1
  :resname o_orderpriority
  :ressortgroupref 1
  :resorigtbl 16561
  :resorigcol 6
  :resjunk false
  }
  {TARGETENTRY
  :expr
  {AGGREF
  :aggfnoid 2803
  :aggtype 20
  :aggcollid 0
  :inputcollid 0
  :aggtranstype 0
  :aggargtypes <>
  :aggdirectargs <>
  :args <>
  :aggorder <>
  :aggdistinct <>
  :aggfilter <>
  :aggstar true
  :aggvariadic false
  :aggkind n
  :aggpresorted false
  :agglevelsup 0
  :aggssplit 0
  :aggno -1
  :aggtransno -1
  :location 32
  }
  :resno 2
  :resname order_count
  :ressortgroupref 0
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
  }
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause {
  {SORTGROUPCLAUSE
  :tleSortGroupRef 1
  :eqop 98
  :sortop 664
  :reverse_sort false
  :nulls_first false
  :hashable true
  }
}
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause {
  {SORTGROUPCLAUSE
  :tleSortGroupRef 1
  :eqop 98
  :sortop 664
  :reverse_sort false
  :nulls_first false
  :hashable true
  }
}
:limitOffset <>

```

```

:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 414
}

```

WARNING: Log 2 : After pull up At starting of Path Generation:

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTRY
    :alias
      {ALIAS
        :aliasname c
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname c
        :colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "
          c_acctbal" "c_mktsegment" "c_comment")
      }
    :rtekind 0
    :relid 16546
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 16546
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
  {FROMEXPR
    :fromlist (
      {RANGETBLREF
        :rtindex 1
      }
    )
  }
  :quals (
    {OEXPR
      :opno 96
      :opfuncid 65
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
          :varno 1
          :varattno 1
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 1
          :location -1
        }
        {PARAM
          :paramkind 1
          :paramid 0
          :paramtype 23
          :paramtypmod -1
          :paramcollid 0
          :location -1
        }
      )
      :location -1
    }
    {OEXPR
      :opno 674
      :opfuncid 297
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
          :varno 1
          :varattno 6
          :vartype 701
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 6
          :location -1
        }
        {CONST
          :consttype 701
          :consttypmod -1
          :constcollid 0
          :constlen 8
          :constbyval true
          :constisnull false
          :location -1
          :constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]
        }
      )
      :location -1
    }
  )
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>

```

```

:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

```

WARNING: Log 3: At the starting of Plan creation:

```

DETAIL: {PATH
:pathtype 339
:parent_relds (b 1)
:pathtarget
{PATHTARGET
:exprs <>
:sortgrouprefs ()
:cost.startup 0
:cost.per_tuple 0
:width 0
:has_volatile_expr 0
}
:required_outer (b)
:parallel_aware false
:parallel_safe false
:parallel_workers 0
:rows 1
:disabled_nodes 0
:startup_cost 0
:total_cost 5922
:pathkeys <>
}

```

WARNING: Log 2 : After pull up At starting of Path Generation:

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:table (
{RANGETBLENTRY
:alias
{ALIAS
:aliasname c
:colnames <>
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "
c_acctbal" "c_mktsegment" "c_comment")
}
:rtekind 0
:relid 16546
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromC1 true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtrindex 1
}
)
:quals (
{OPEXPR
:opno 674
:opfncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]
}
)
:location -1
}
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY

```

```

:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

```

WARNING: Log 3: At the starting of Plan creation:

```

DETAIL: {PATH
:pathtype 339
:parent_relds (b 1)
:pathtarget
{PATHTARGET
:exprs (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
)
:sortgrouprefs ( 0)
:cost.startup 0
:cost.per_tuple 0
:width 4
:has_volatile_expr 0
}
:required_outer (b)
:parallel_aware false
:parallel_safe true
:parallel_workers 0
:rows 67000
:disabled_nodes 0
:startup_cost 0
:total_cost 5547
:pathkeys <>
}

```

WARNING: Log 2 : After pull up At starting of Path Generation:

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity true
:hasGroupRTE true
:isReturn false
:cteList <>
:table (
{RANGEBLENTTRY
:alias <>
:eref
{ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 16561
:inh false
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals ((
{ALTERNATIVESUBPLAN
:subplans (
{SUBPLAN
:subLinkType 0
:testexpr <>
:paramids <>
:plan_id 1
:plan_name SubPlan\ 1
:firstColType 2270
:firstColTypmod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe false
:setParam <>
:parParam (i 0)
:args (
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1

```

```

        :varattnosyn 2
        :location -1
    }
)
:startup_cost 0
:per_call_cost 5922
}
{SUBPLAN
:sublinkType 2
:testexpr
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 2
:varatype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
{PARAM
:paramkind 1
:paramid 1
:paramtype 23
:paramitypmode -1
:paramcollid 0
:location -1
}
)
:location -1
}
:params (i 1)
:plan_id 2
:plan_name SubPlan\ 2
:firstColType 23
:firstColTypeMode -1
:firstColCollation 0
:useHashTable true
:unknownEqFalse true
:parallel_safe true
:setParam <>
:parParam <>
:args <>
:startup_cost 5716.52
:per_call_cost 0.0025
}
)
}
))
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
:rtekind 9
:groupexprs (
{VAR
:varno 1
:varattno 6
:varatype 25
:varitypmode -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 11
}
)
:lateral false
:inFromCl false
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:permindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16561
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals (
{OEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16

```

```

:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 5
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 5
    :location 82
  }
  {CONST
    :consttype 1082
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 102
    :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
  }
)
:location 94
}
{OPEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 5
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 5
    :location 123
  }
  {CONST
    :consttype 1114
    :consttypmod -1
    :constcollid 0
    :constlen 8
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
  }
)
:location 135
}
)
}
:rarg
{RANGETBLREF
:rtindex 3
}
:usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 3
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 1
    :location 282
  }
  {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 295
  }
)
:location 293
}
{OPEXPR
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 3
    :varattno 12
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 12
    :location 322
  }
  {VAR
    :varno 3
    :varattno 13
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 13
    :location 337
  }
)
:location 335
}
)

```

```

        :alias <>
        :rtindex 0
    }
}
:quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
    :expr
    {VAR
    :varno 1
    :varattno 6
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 6
    :location -1
    }
    :resno 1
    :resname o_orderpriority
    :ressortgroupref 1
    :resorigtbl 16561
    :resorigcol 6
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {AGGREF
    :aggfnoid 2803
    :aggtype 28
    :aggcollid 0
    :inputcollid 0
    :aggtranstype 0
    :aggargtypes <>
    :aggdirectargs <>
    :args <>
    :aggorder <>
    :aggdistinct <>
    :aggfilter <>
    :aggstar true
    :aggvariadic false
    :aggkind n
    :aggpresorted false
    :agglevelsup 0
    :aggssplit 0
    :aggno -1
    :aggtransno -1
    :location 32
    }
    :resno 2
    :resname order_count
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    )
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
    {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
    }
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
    {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
    }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 414
}

WARNING:  Log 3: At the starting of Plan creation:
DETAIL:   {AGGPATH
:path.pathtype 365
:parent_relds (b)
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe false
:path.parallel_workers 0
:path.rows 5
:path.disabled_nodes 0
:path.startup_cost 8883274415.788885
:path.total_cost 8883274719.763882
:path.pathkeys (
    {PATHKEY
    :pk_eclass
    {EQUIVALENCECLASS
    :ec_opfamilies (o 1994 2095)
    :ec_collation 100
    :ec_childmembers_size 0
    :ec_members (
        {EQUIVALENCEMEMBER
        :em_expr
        {VAR
        :varno 1
        :varattno 6
        :vartype 25
        :vartypmod -1
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 6
        :location -1
        }
        :em_relds (b 1)
        :em_is_const false
        :em_is_child false
        :em_datatype 25
        :em_jdomain
        {JOINDOMAIN
        :jd_relds (b 1 3)
        }
        }
    }
    }
)

```

```

    }
    :ec_childmembers <>
    :ec_sources <>
    :ec_derives_list <>
    :ec_relids (b 1)
    :ec_has_const false
    :ec_has_volatile false
    :ec_broken false
    :ec_sortref 1
    :ec_min_security 4294967295
    :ec_max_security 0
  }
  :pk_opfamily 1994
  :pk_cmptype 1
  :pk_nulls_first false
}

:subpath
{SORTPATH
:path.pathtype 362
:parent_relids (b)
:path.pathtarget
{PATHTARGET
:exprs (
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
)
:sortgrouprefs ( 1)
:cost.startup 0
:cost.per_tuple 0
:width 9
:has_volatile_expr 0
}
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe false
:path.parallel_workers 0
:path.rows 40534
:path.disabled_nodes 0
:path.startup_cost 8883274415.708885
:path.total_cost 8883274517.043884
:path.pathkeys (
{PATHKEY
:pk_eclass
{EQUIVALENCECLASS
:ec_opfamilies (o 1994 2095)
:ec_collation 100
:ec_childmembers_size 0
:ec_members (
{EQUIVALENCEMEMBER
:em_expr
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:em_relids (b 1)
:em_is_const false
:em_is_child false
:em_datatype 25
:em_jdomain
{JOINDOMAIN
:jd_relids (b 1 3)
}
}
)
:ec_childmembers <>
:ec_sources <>
:ec_derives_list <>
:ec_relids (b 1)
:ec_has_const false
:ec_has_volatile false
:ec_broken false
:ec_sortref 1
:ec_min_security 4294967295
:ec_max_security 0
}
:pk_opfamily 1994
:pk_cmptype 1
:pk_nulls_first false
}

:subpath
{HASHPATH
:jpath.path.pathtype 359
:parent_relids (b 1 3)
:jpath.path.pathtarget
{PATHTARGET
:exprs (
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
)
:sortgrouprefs ( 1)
:cost.startup 0
:cost.per_tuple 0
:width 9
:has_volatile_expr 0
}
:required_outer (b)
:jpath.path.parallel_aware false
:jpath.path.parallel_safe false
:jpath.path.parallel_workers 0
:jpath.path.rows 40534
:jpath.path.disabled_nodes 0
:jpath.path.startup_cost 212549.3375
:jpath.path.total_cost 8883271313.470623
:jpath.path.pathkeys <>
:jpath.jointype 4
:jpath.inner_unique false
:jpath.outerjoinpath
{PATH
:pathtype 339
:parent_relids (b 1)
:required_outer (b)
:parallel_aware false
:parallel_safe false
:parallel_workers 0

```

```

:rows 144199
:disabled_nodes 0
:startup_cost 0
:total_cost 8883048508
:pathkeys <>
}
:jpath.innerjoinpath
{PATH
:pathtype 339
:parent_relids (b 3)
:required_outer (b)
:parallel_aware false
:parallel_safe true
:parallel_workers 0
:rows 2000307
:disabled_nodes 0
:startup_cost 0
:total_cost 179731.5
:pathkeys <>
}
:jpath.joinrestrictinfo (
{RESTRICTINFO
:clause
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 295
}
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 282
}
}
:location -1
}
:is_pushed_down true
:can_join true
:pseudoconstant false
:has_clone false
:is_clone false
:leakproof true
:has_volatile 2
:security_level 1
:num_base_rels 2
:clause_relids (b 1 3)
:required_relids (b 1 3)
:incompatible_relids (b)
:outer_relids (b)
:left_relids (b 1)
:right_relids (b 3)
:orclause <>
:rinfo.serial 6
:eval_cost.startup 0
:eval_cost.per_tuple 0.0025
:norm_selec 6.666666666666667e-07
:outer_selec 0.28109466666666666
:mergeopfamilies (o 1976)
:left_em
{EQUIVALENCEMEMBER
:em_expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 295
}
:em_relids (b 1)
:em_is_const false
:em_is_child false
:em_datatype 23
:em_jdomain
{JOINDOMAIN
:jd_relids (b 1 3)
}
}
:right_em
{EQUIVALENCEMEMBER
:em_expr
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 282
}
:em_relids (b 3)
:em_is_const false
:em_is_child false
:em_datatype 23
:em_jdomain
{JOINDOMAIN
:jd_relids (b 1 3)
}
}
:outer_is_left false
:hashjoinoperator 96
:left_bucketsize 6.934860852017004e-06
:right_bucketsize 7.115057596391243e-06
:left_mcvfreq 0
:right_mcvfreq 0
:left_hasheqoperator 96
:right_hasheqoperator 96
}
}
:path_hashclauses (
{RESTRICTINFO
:clause
{OEXPR
:opno 96
:opfuncid 65

```

```

:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 295
  }
  {VAR
    :varno 3
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 1
    :location 282
  }
)
:location -1
}
:is_pushed_down true
:can_join true
:pseudoconstant false
:has_clone false
:is_clone false
:leakproof true
:has_volatile 2
:security_level 1
:num_base_rels 2
:clause_relds (b 1 3)
:required_relds (b 1 3)
:incompatible_relds (b)
:outer_relds (b)
:left_relds (b 1)
:right_relds (b 3)
:orclause <
:rinfo_serial 6
:eval_cost.startup 0
:eval_cost.per_tuple 0.0025
:norm_selec 6.666666666666667e-07
:outer_selec 0.28109466666666666
:mergeopfamilies (o 1976)
:left_em
{EQUIVALENCEMEMBER
  :em_expr
  {VAR
    :varno 1
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 295
  }
  :em_relds (b 1)
  :em_is_const false
  :em_is_child false
  :em_datatype 23
  :em_jdomain
  {JOINDOMAIN
    :jd_relds (b 1 3)
  }
}
:right_em
{EQUIVALENCEMEMBER
  :em_expr
  {VAR
    :varno 3
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 1
    :location 282
  }
  :em_relds (b 3)
  :em_is_const false
  :em_is_child false
  :em_datatype 23
  :em_jdomain
  {JOINDOMAIN
    :jd_relds (b 1 3)
  }
}
:outer_is_left false
:hashjoinoperator 96
:left_bucketsize 6.934860852017004e-06
:right_bucketsize 7.115057596391243e-06
:left_mcvfreq 0
:right_mcvfreq 0
:left_hasheqoperator 96
:right_hasheqoperator 96
}
:num_batches 16
:inner_rows_total 2000307
}
}
:aggstrategy 1
:aggsplit 0
:numGroups 5
:transitionSpace 0
:groupClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:qual <
}

```

WARNING: Log 4: At the Executor starting point we have proposed Plan by Planner:

```

DETAIL: {PLANNEDSTMT
:commandType 1
:queryId 0
:planId 0
:hasReturning false
:hasModifyingCTE false
:canSetTag true
:transientPlan false
:dependsOnRole false
:parallelModeNeeded false

```

```

:jitFlags 31
:planTree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 8883274415.708885
:plan.total_cost 8883274719.763882
:plan.plan_rows 5
:plan.plan_width 17
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 0
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 0
:varattnosyn 0
:location -1
}
:resno 1
:resname o_orderpriority
:ressortgroupref 1
:resorigtbl 16561
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 20
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 0
:aggno 0
:aggtransno 0
:location 32
}
:resno 2
:resname order_count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:plan.qual <>
:plan.lefttree
{SORT
:plan.disabled_nodes 0
:plan.startup_cost 8883274415.708885
:plan.total_cost 8883274517.043884
:plan.plan_rows 40534
:plan.plan_width 9
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 1
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 1
:resname <>
:ressortgroupref 1
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:plan.qual <>
:plan.lefttree
{HASHJOIN
:join.plan.disabled_nodes 0
:join.plan.startup_cost 212549.3375
:join.plan.total_cost 8883271313.470623
:join.plan.plan_rows 40534
:join.plan.plan_width 9
:join.plan.parallel_aware false
:join.plan.parallel_safe false
:join.plan.async_capable false
:join.plan.plan_node_id 2
:join.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 1
:resname <>
:ressortgroupref 1
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:join.plan.qual <>
:join.plan.lefttree
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 8883048508
:scan.plan.plan_rows 144199
:scan.plan.plan_width 13
:scan.plan.parallel_aware false

```

```

:scan.plan.parallel_safe false
:scan.plan.async_capable false
:scan.plan.plan_node_id 3
:scan.plan.targetlist (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 6
      :vartype 25
      :vartypmod -1
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 6
      :location -1
    }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location 295
    }
    :resno 2
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:scan.plan.qual (
  {OPEXPR
    :opno 1098
    :opfuncid 1090
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 5
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 5
        :location 82
      }
      {CONST
        :consttype 1082
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location 102
        :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
      }
    )
    :location 94
  }
  {SUBPLAN
    :subLinkType 2
    :testexpr
    {OPEXPR
      :opno 96
      :opfuncid 65
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
          :varno 1
          :varattno 2
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 2
          :location -1
        }
        {PARAM
          :paramkind 1
          :paramid 1
          :paramtype 23
          :paramtypmod -1
          :paramcollid 0
          :location -1
        }
      )
      :location -1
    }
  }
  :paramIds (i 1)
  :plan_id 2
  :plan_name SubPlan\ 2
  :firstColType 23
  :firstColTypmod -1
  :firstColCollation 0
  :useHashTable true
  :unknownEqFalse true
  :parallel_safe true
  :setParam <>
  :parParam <>
  :args <>
  :startup_cost 5716.52
  :per_call_cost 0.0025
)
{OPEXPR
  :opno 2345
  :opfuncid 2338
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
      :varno 1
      :varattno 5
      :vartype 1082
      :vartypmod -1
      :varcollid 0
    }
  )
}

```

```

:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 123
}
(CONST
:consttype 1114
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
)
}
:location 135
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.scanreliid 1
}
:join.plan.righttree
{HASH
:plan.disabled_nodes 0
:plan.startup_cost 179731.5
:plan.total_cost 179731.5
:plan.plan_rows 2000307
:plan.plan_width 4
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 4
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 179731.5
:scan.plan.plan_rows 2000307
:scan.plan.plan_width 4
:scan.plan.parallel_aware false
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 5
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 282
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:scan.plan.qual (
{OPEXPR
:opno 1095
:opfuncid 1007
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 12
:location 322
}
{VAR
:varno 3
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 13
:location 337
}
)
:location 335
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.scanreliid 3
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b)

```

```

:plan.allParam (b)
:hashkeys (
  {VAR
    :varno -2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 1
    :location 282
  }
)
:skewTable 16561
:skewColumn 1
:skewInherit false
:rows_total 0
}
:join.plan.initPlan <>
:join.plan.extParam (b)
:join.plan.allParam (b)
:join.jointype 4
:join.inner_unique false
:join.joinqual <>
:hashclauses (
  {OEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno -2
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location 295
      }
      {VAR
        :varno -1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 1
        :location 282
      }
    )
  }
  :location -1
}
)
:hashoperators (o 96)
:hashcollations (o 0)
:hashkeys (
  {VAR
    :varno -2
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 1
    :location 295
  }
)
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b)
:plan.allParam (b)
:numCols 1
:sortColIdx ( 1)
:sortOperators ( 664)
:collations ( 100)
:nullsFirst ( false)
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b)
:plan.allParam (b)
:aggstrategy 1
:aggssplit 0
:numCols 1
:grpColIdx ( 1)
:grpOperators ( 98)
:grpCollations ( 100)
:numGroups 5
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:partPruneInfos <>
:rtable (
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname orders
        :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
      }
    :rtekind 0
    :relid 16561
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname *GROUP*
        :colnames ("o_orderpriority")
      }
    :rtekind 9
    :groupexprs <>
    :lateral false
    :inFromCl false
    :securityQuals <>
  }
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS

```

```

:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname c
:colnames <>
}
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")
}
:rtekind 0
:relid 16546
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname c
:colnames <>
}
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")
}
:rtekind 0
:relid 16546
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:unprunableRelids (b 1 3 4 5)
:permInfos (
{RTEPERMISSIONINFO
:relid 16561
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
}
:resultRelations <>
:appendRelations <>
:subplans (<>
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 5547
:scan.plan.plan_rows 67888
:scan.plan.plan_width 4
:scan.plan.parallel_aware false
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 6
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 5
:varattno 1
:varatype 23
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
)
:scan.plan.qual (
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0

```

```

:args (
  {VAR
    :varno 5
    :varattno 6
    :vartype 701
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 5
    :varattnosyn 6
    :location -1
  })
  {CONST
    :consttype 701
    :consttypmod -1
    :constcollid 0
    :constlen 8
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]
  })
  :location -1
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.scanreliid 5
)
:rewindPlanIDs (b)
:rowMarks <>
:relationOids (o 16561 16566 16546 16546)
:invalidItems <>
:paramExecTypes (o 23 23)
:utilityStmt <>
:stmt_location 0
:stmt_len 414
}

```

o_orderpriority	order_count
1-URGENT	4751
2-HIGH	4737
3-MEDIUM	4661
4-NOT SPECIFIED	4591
5-LOW	4577
(5 rows)	

tpch=>